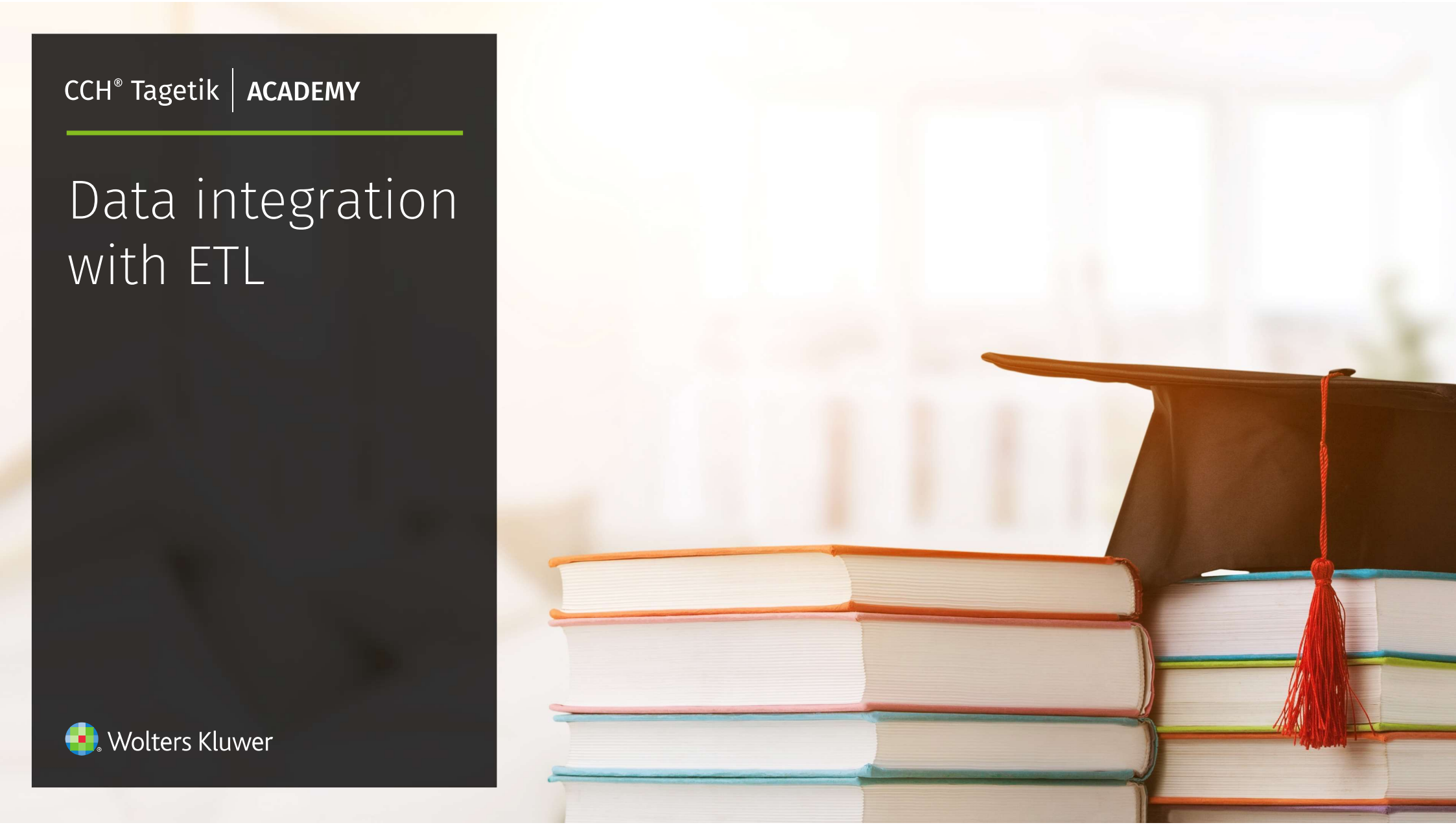
CCH® Tagetik | ACADEMY

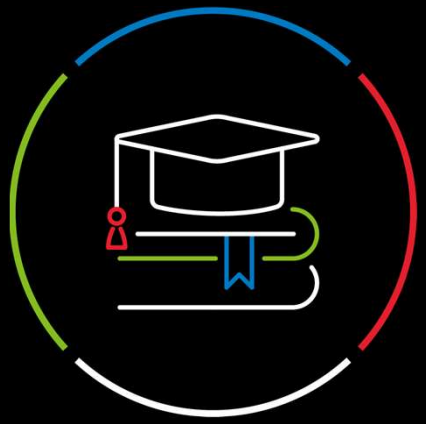
Data integration with ETL



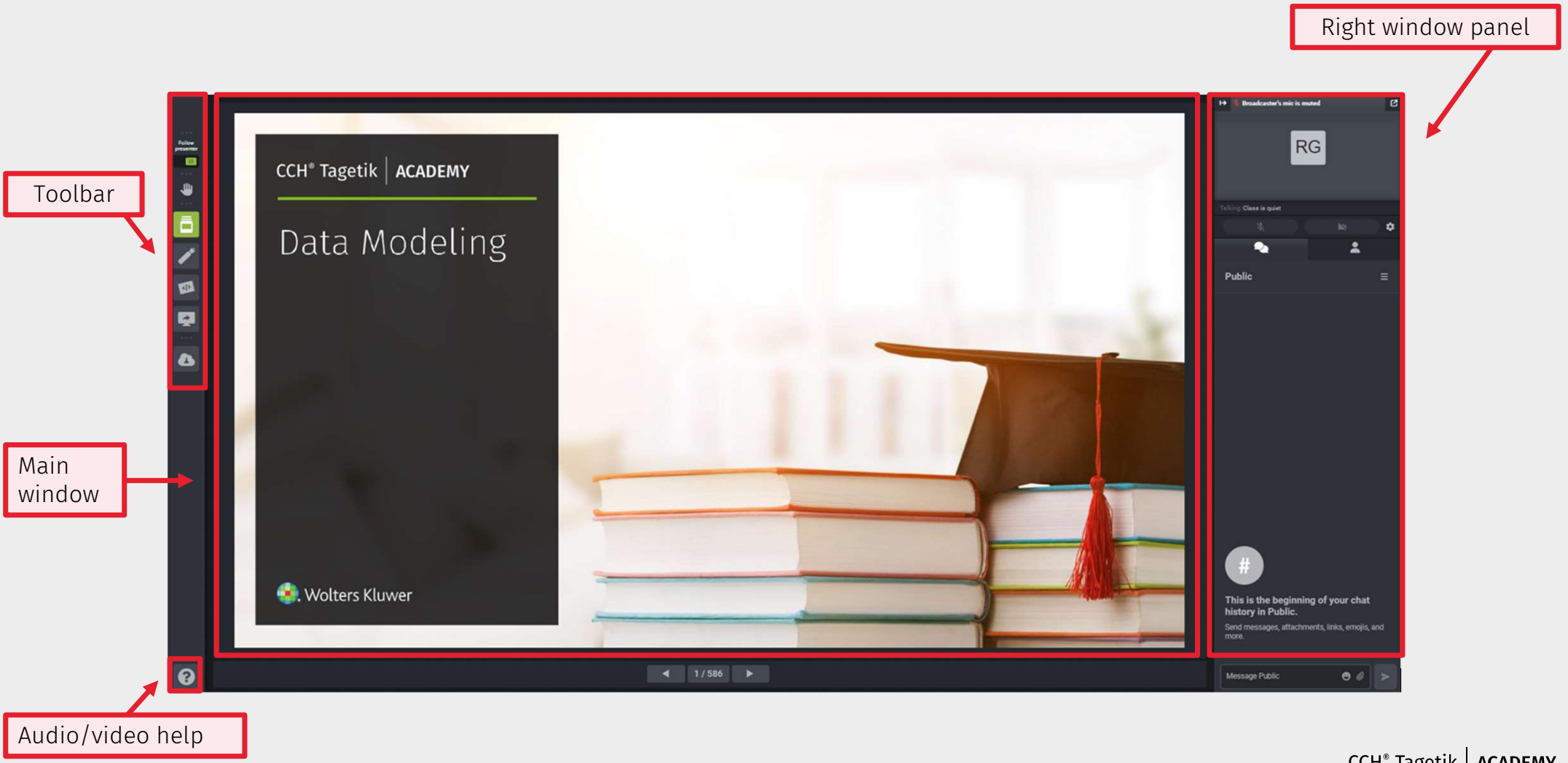
Wolters Kluwer

Before starting

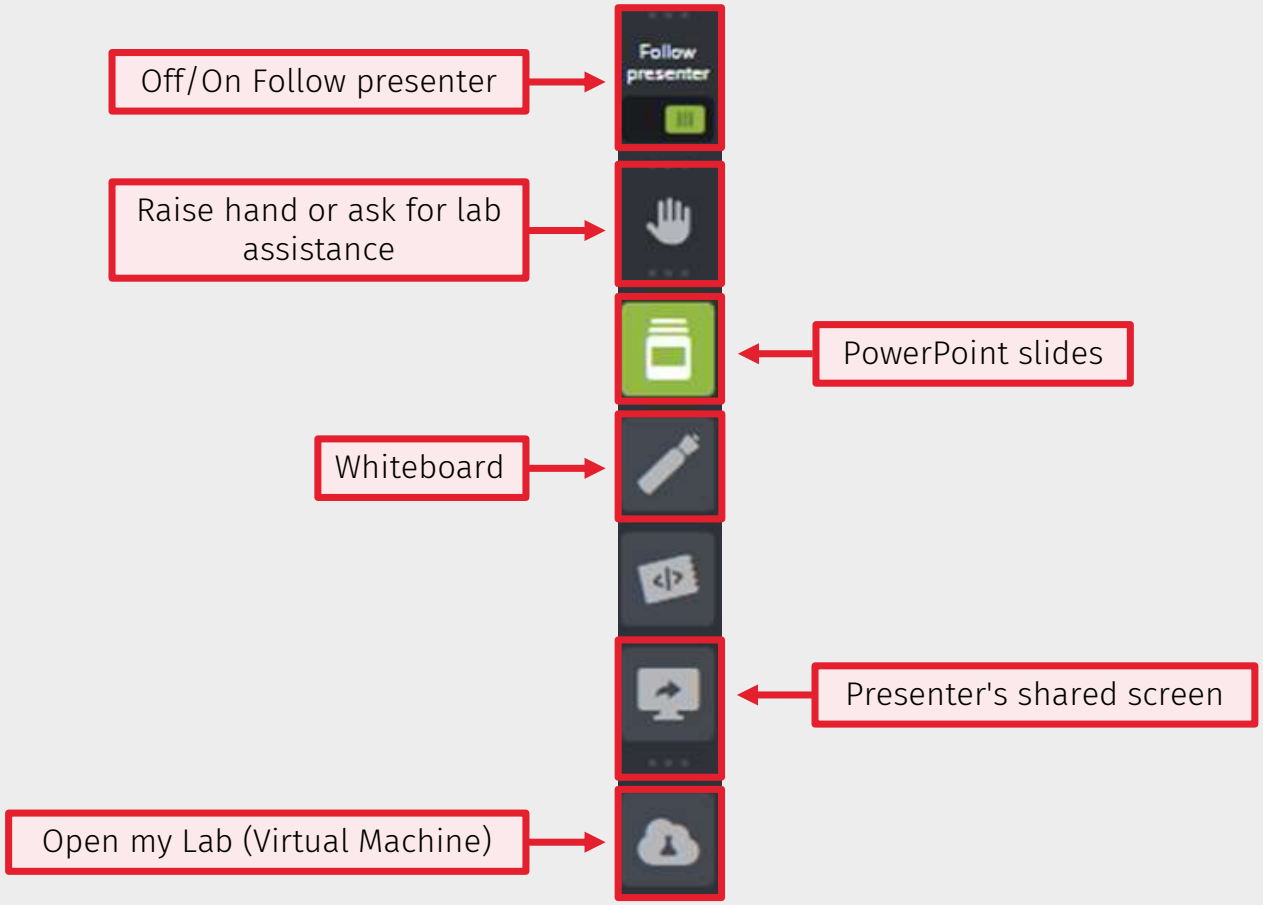
Strigo platform - “How to”



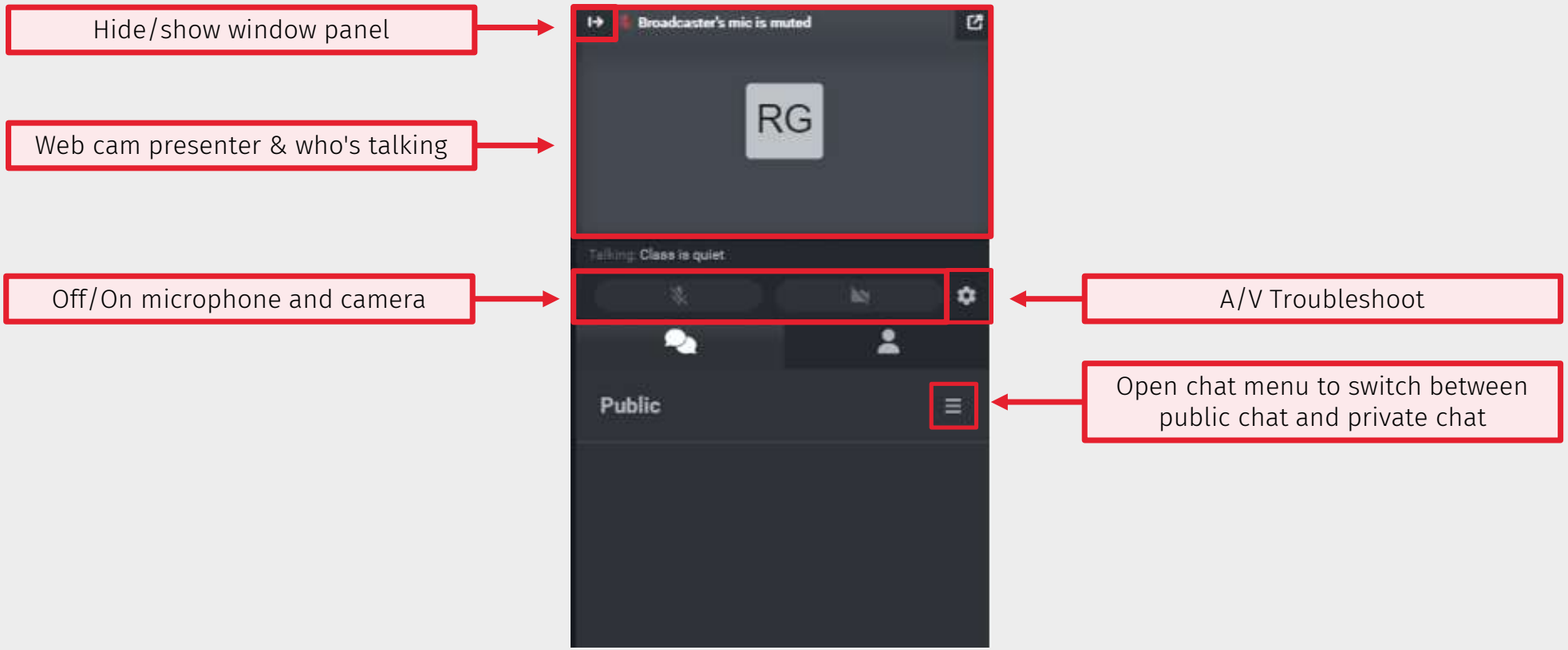
Strigo overview (for Instructor-led classes)



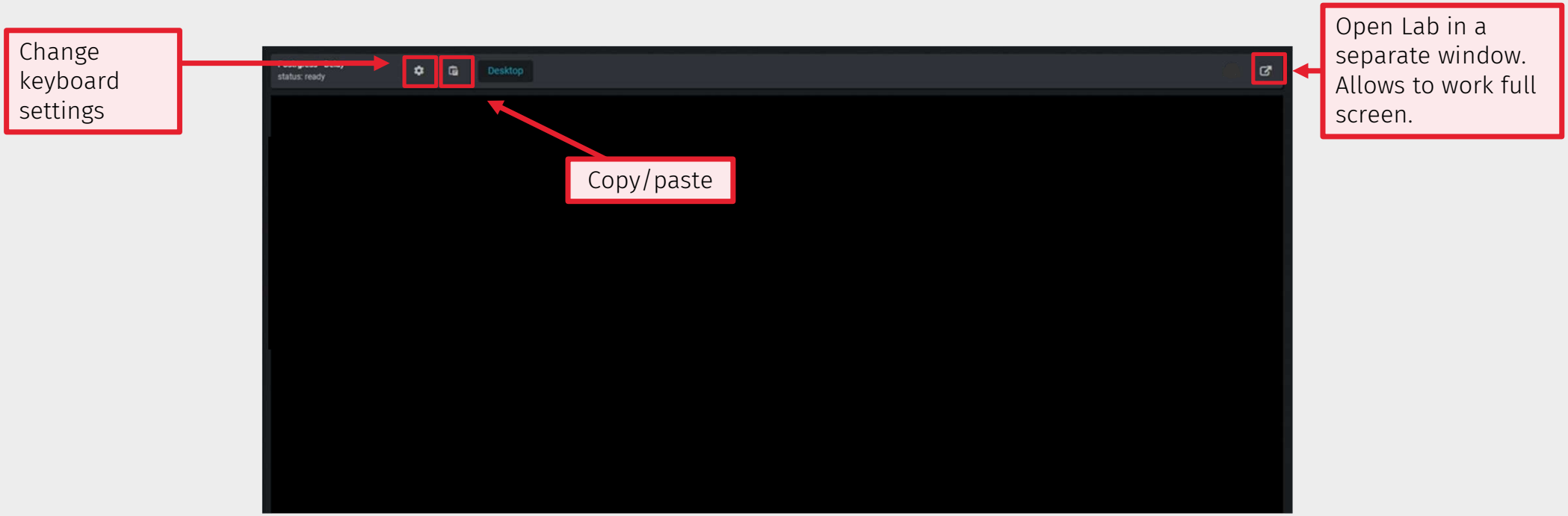
Strigo left toolbar



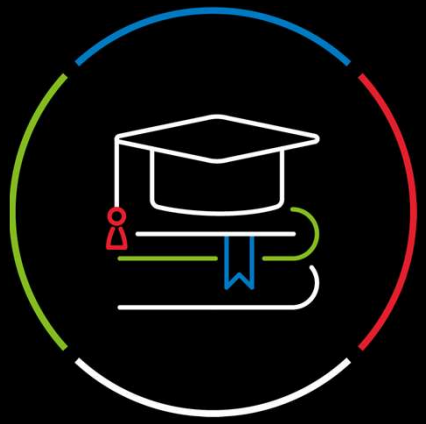
Strigo right window panel



Strigo Lab (Virtual Machine)

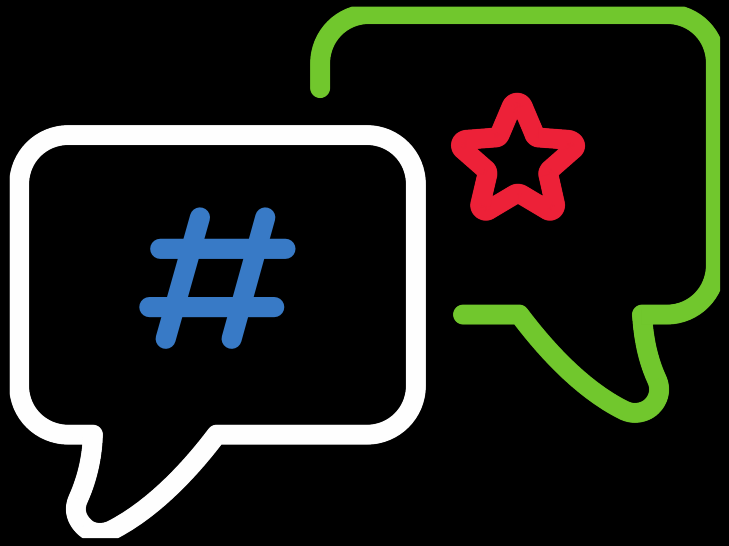


Agenda



- ETL Terminology
- Architecture
- Configuration
- Dimensions and Parameters
- Data sources
- Transformations
- Loadings
- Routine
- Job
- Troubleshooting and analysis

Let’s talk about: ETL features



CCH Tagetik ETL

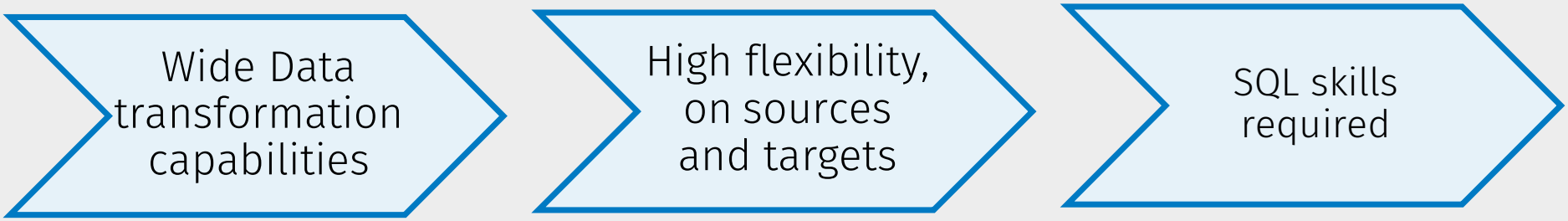
ETL is a mature and extremely reliable Tool that export/import financial data and metadata even with high data volume

It works as an interface to Tagetik -> Tagetik open architecture

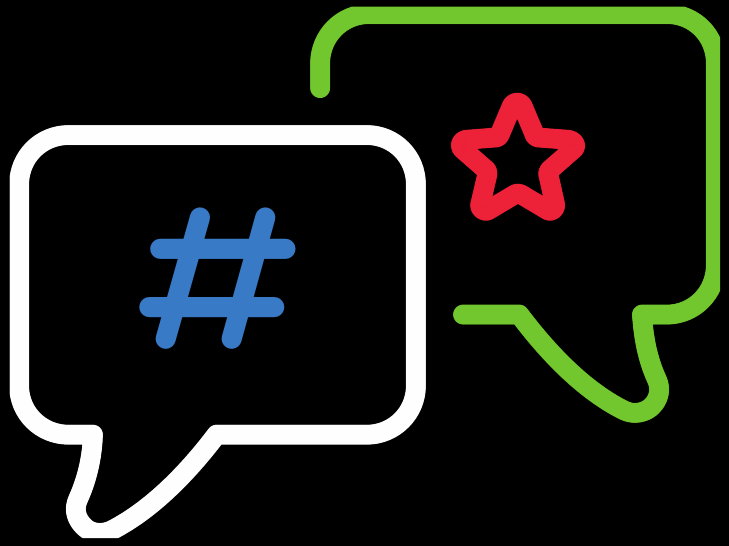
Ensure security of data and error handling (Instead of loading directly on a table)

Fast development/Out of the box functionalities

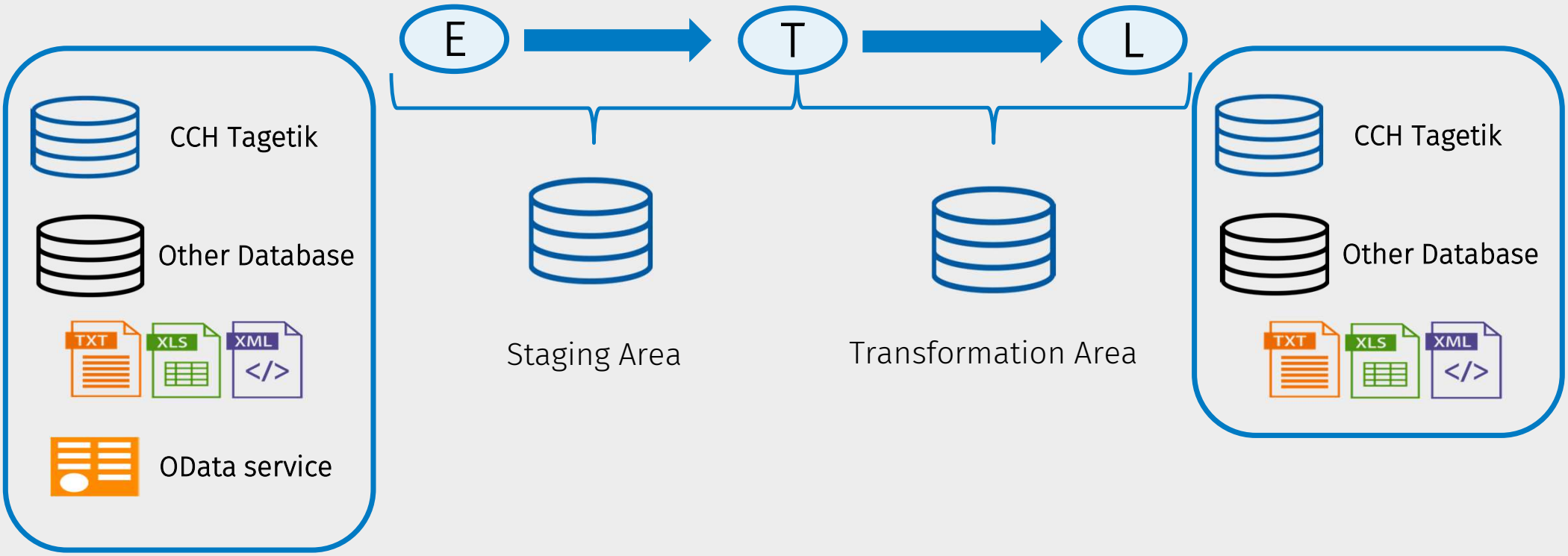
ETL engine


Wide Data transformation capabilities
High flexibility, on sources and targets
SQL skills required

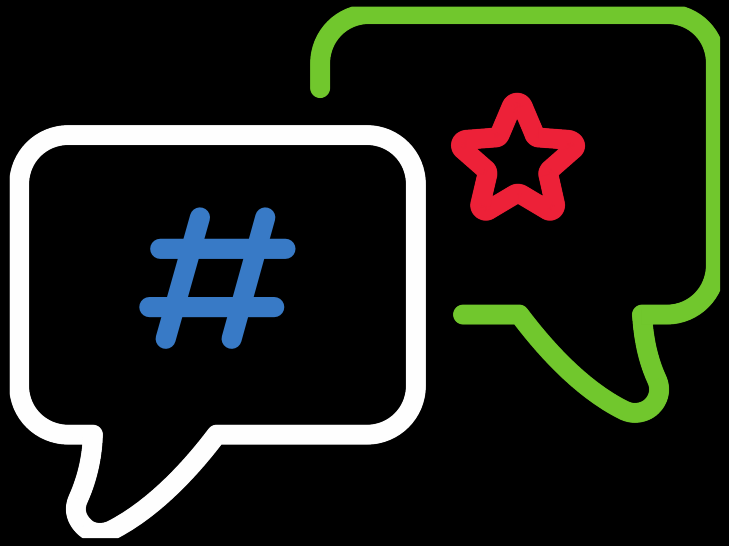
Let’s talk about: ETL Architecture



ETL Architecture

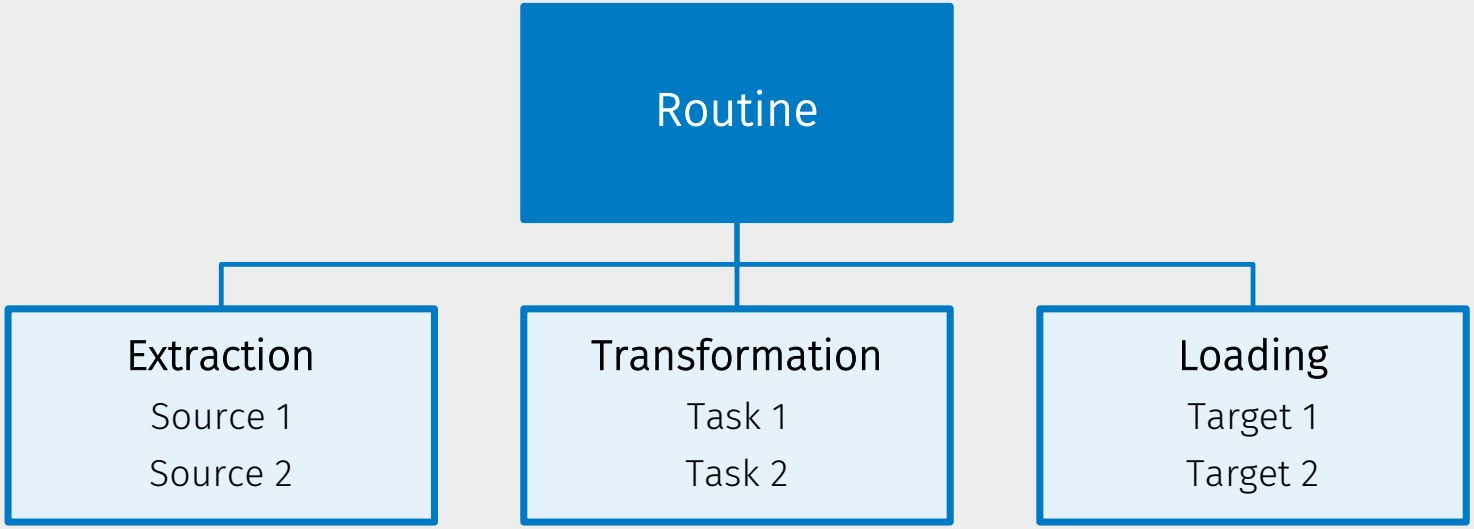


Let’s talk about: ETL Terminology



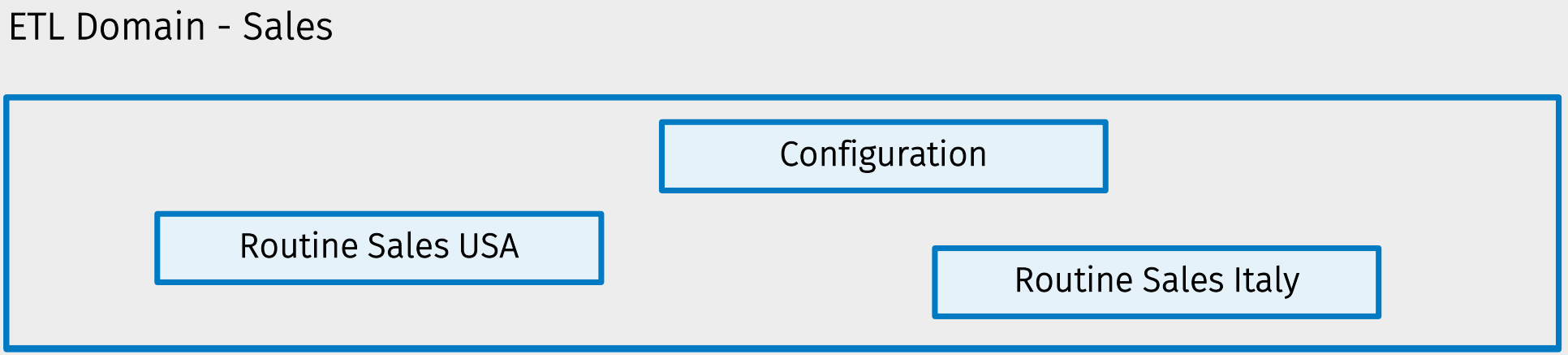
ETL Routine

Data processing smallest unit of ETL Process.
Combine details for data extraction, transformation and loading.



ETL Domain

Macro container for all settings, parameters, routines and lookup tables



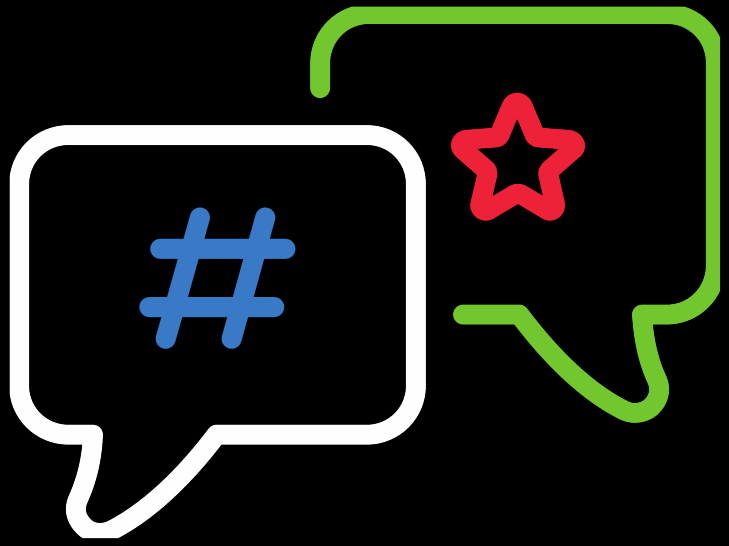
ETL job

Allows to run in sequence several routines belonging to different domains

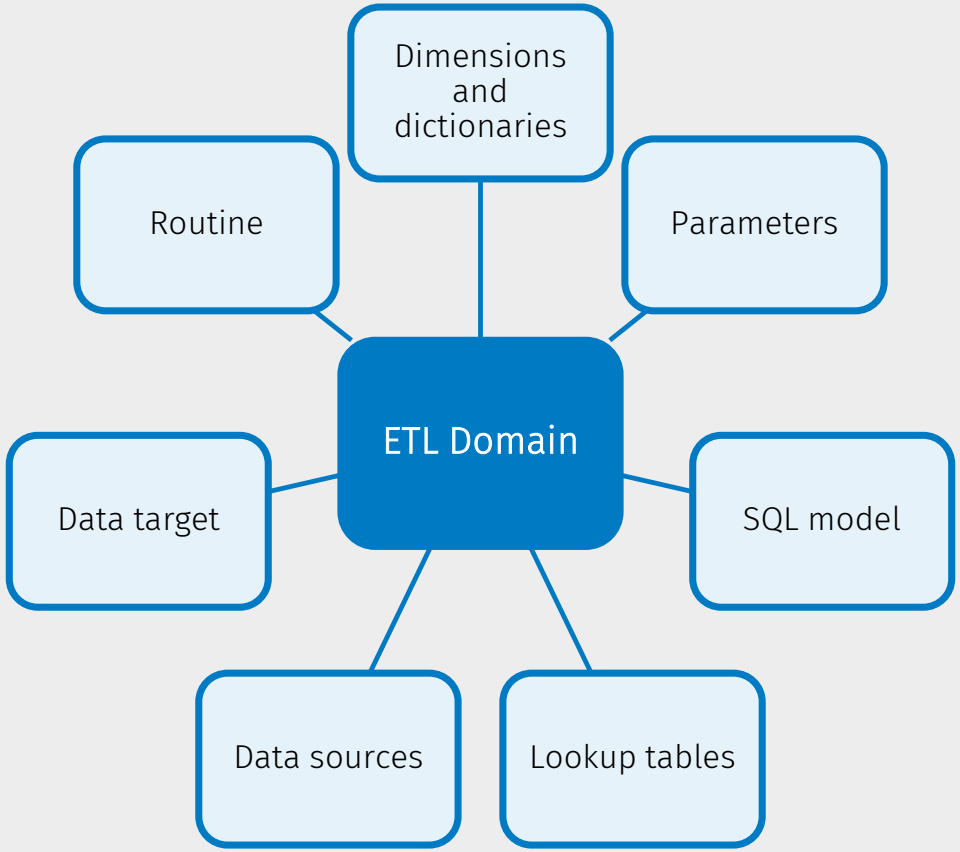
Executable on demand in different ways and on a scheduled basis

Preferred way to run ETL by end users

Let’s talk about: Configuration



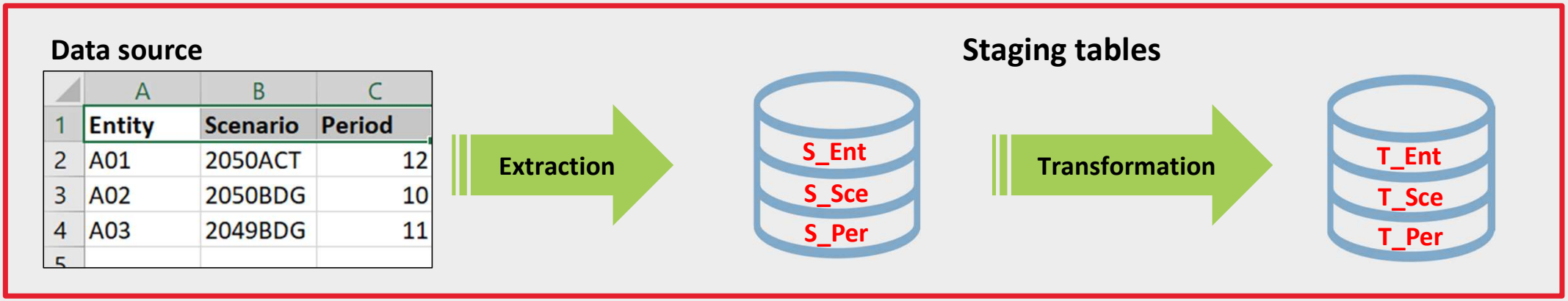
ETL Definition - Structure



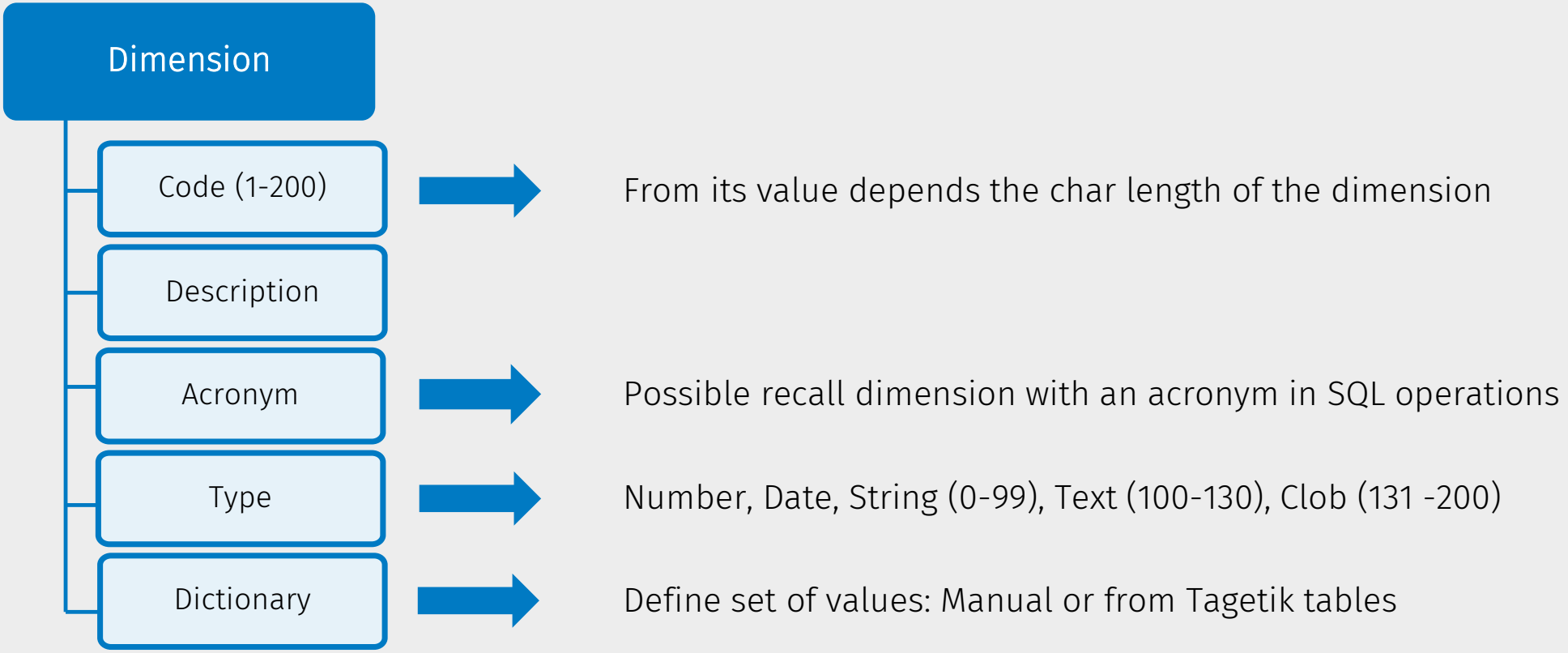
- ETL components belongs to the domain
- Creation of the routine is the last step of configuration

Staging table fields/Dimensions

- Staging table fields (STF) are a fundamental ingredient for any ETL routine definition
- Represent columns of the staging table where Tagetik stores extracted and transformed data

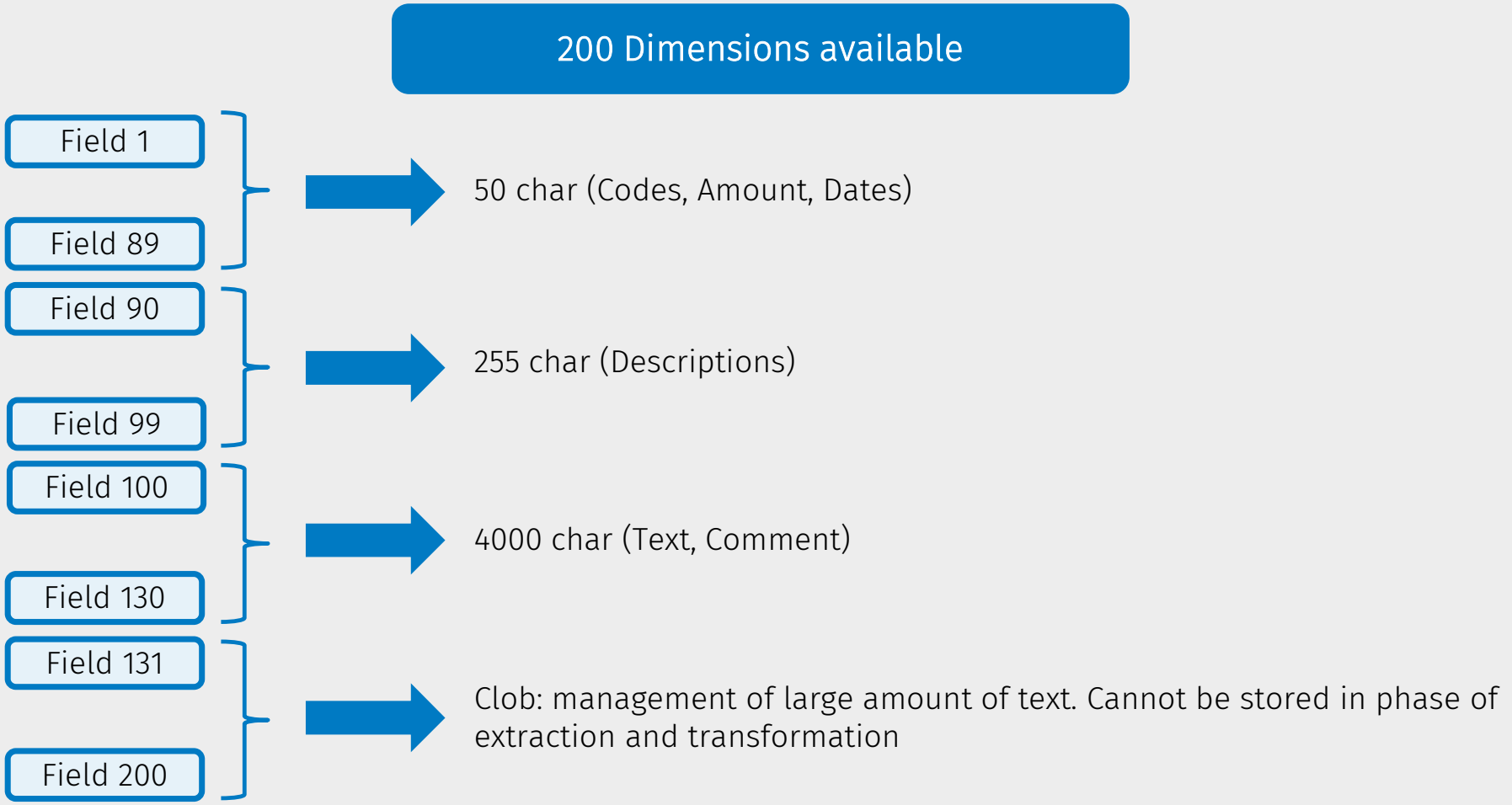


Dimension definition



Dimensions are referred in all components of the domain

Staging table fields/Dimensions



Routine - Data log tables

While running an ETL, data extracted and transformed can be stored in tables
Dimensions Column name is given from **CAMPO + 'Code Field'**

MAP_DATI_ACQUISITI

COD_UTENTE	COD_MAPPATURA	COD_IMPORT	CAMPO1	...	CAMPO130
Wile	ETL_DOMAIN_1	ROUTINE_01	2049		100
Wile	ETL_DOMAIN_1	ROUTINE_01	2049		200

Holds data extracted

MAP_DATI_TRASFORMATI

COD_MAPPATURA	COD_IMPORT	CAMPO1	CAMPO2	...	CAMPO130
ETL_DOMAIN_2	ROUTINE_02	2049ACT	$AMOUNT		100
ETL_DOMAIN_2	ROUTINE_02	2049ACT	$AMOUNT		300

Holds data transformed

Parameters

Parameters in Extraction/Transformation

- Deletion criteria in staging tables
- Extraction/transformation filter from source to staging table

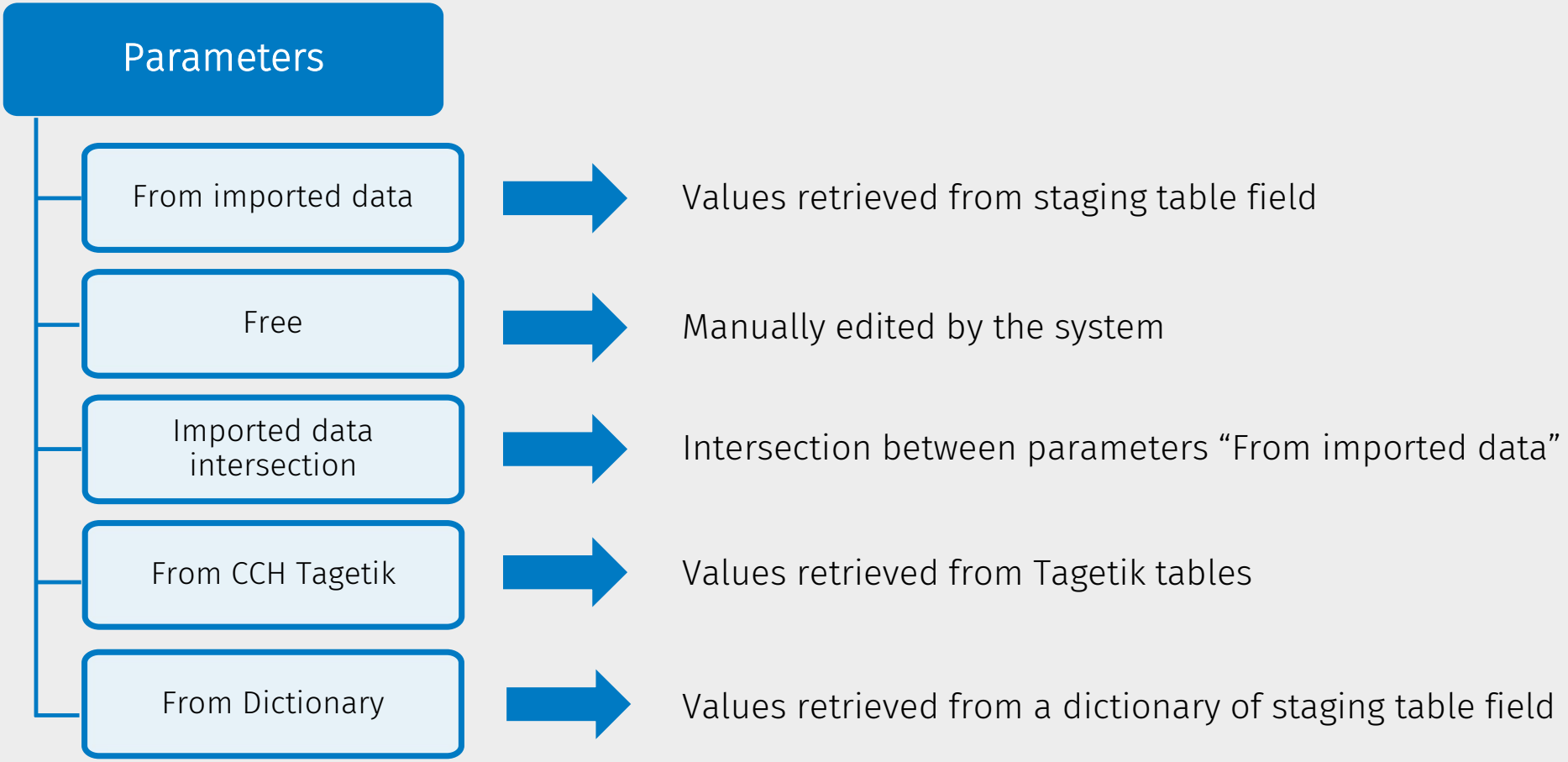
Parameters in Loading

- Deletion criteria in Tagetik table/Target
- Loading filter from Staging table to Target

Transformation Parameter choice

- Used as custom transformation
- Limit the amount of data to transform
- Assign to a dimension a fixed value

Parameters - Definition



Parameters – Data intersection

2 single parameters:

```
Par_Scenario = 2050BDG
Par_Period   = 12
```

- OR selection

S_SCENARIO	S_PERIOD	S_AMOUNT
2050BDG	06	1000
2050BDG	12	2000
2049ACT	12	3000
2049ACT	06	4000

Data intersection parameter:

```
Par_Scenario/Period = 2050BDG/12
```

- AND selection

S_SCENARIO	S_PERIOD	S_AMOUNT
2050BDG	06	1000
2050BDG	12	2000
2049ACT	12	3000
2049ACT	06	4000

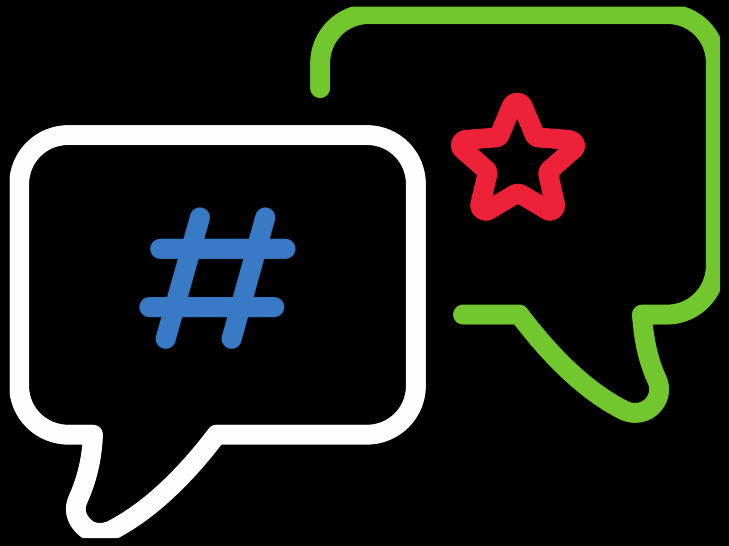
Now it’s your turn

Exercise # 1

Setup staging table and parameters

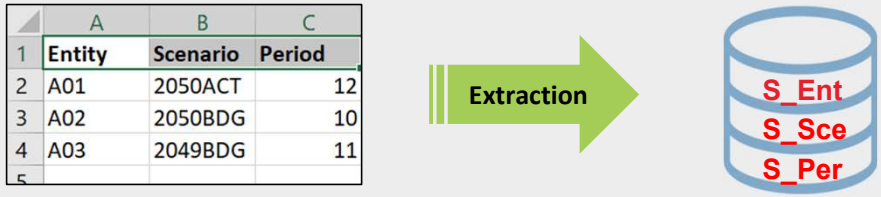


Let’s talk about: Extraction

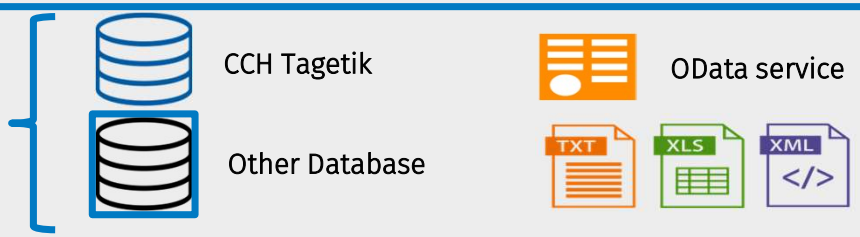


Data Sources

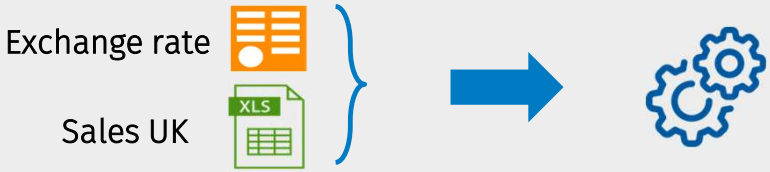
Each data source holds the mapping between its fields and STF



Data source can extract data from



In Extraction, a routine can retrieve data from multiple sources



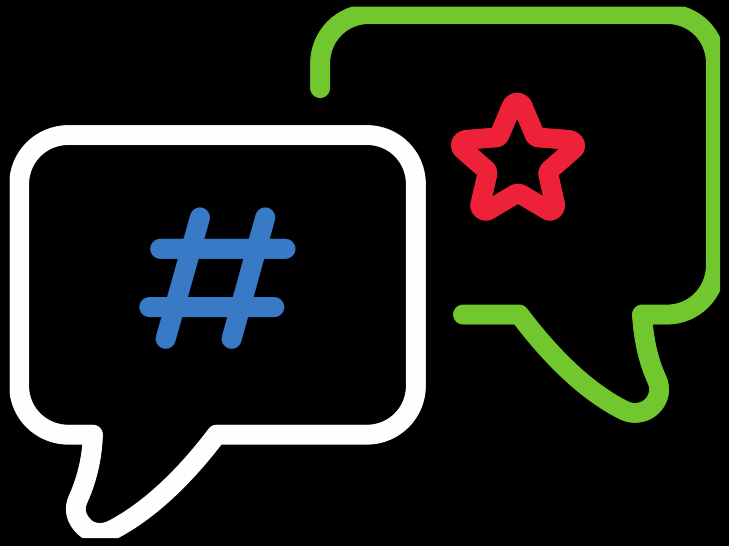
Now it’s your turn

Exercise # 2

Setup Extraction



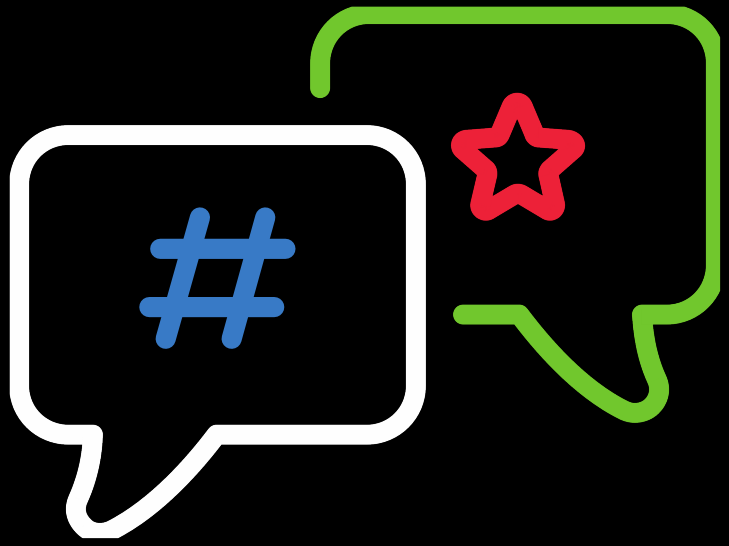
Let’s talk about: Transformations



Transformations

- Set the data extracted compliant with target
- System provides many tools/operations
- Lookup Tables are general tranformations available at domain level
- Other Transformations are defined ad-hoc in phase of routine
- SQL knowledge is required

Let’s talk about: Lookup tables



Lookup tables

- Perform mapping between dimensions of Staging table fields
- Can be defined more input fields and more output fields
- It is a component of the domain. Can be used from different domain's routine
- Used by some transformation tasks in the routine
- Easy to handle by business administrator

Input 1	Input 2	Input N	Output 1	Output 2	Output N
In_1	In_2	In_N	Out_1	Out_2	Out_N
In_3	In_4	In_X	Out_3	Out_4	Out_X

Lookup tables – Example 1:1

Account in General Ledger can have different codes than Tagetik account.
A mapping is often required.

ERP ACCOUNT (Input)	TAGETIK ACCOUNT (Output)
X01 – Revenue ERP1	501 – Gross Revenues
X02 – Revenue ERP2	501 – Gross Revenues
X03 - Revenue ERP3	502 – Other Revenues

Lookup Tables – Example N:1

Cost centers in General Ledger use different account structures. The same account numbers from different cost centers are mapped to different Tagetik accounts.

COST CENTER (Input)	ERP ACCOUNT (Input)	TGK ACCOUNT (Output)
A01 – Sales	X01 – Revenue ERP	501 – Gross revenues
A02 – Sales UK	X01 – Revenue ERP	502 – Other revenues
A03 - Costs	C03 – Cost ERP	600 - Material costs

Lookup tables: wildcard char (%)

Use of SQL like operator **“%“** can be enable with an option

Example defining **“X%“** means **“Whatever starts with X”**

ERP ACCOUNT (Input)	TGK ACCOUNT (Output)
X%	501 – Gross revenues
Y%	601 – Production costs

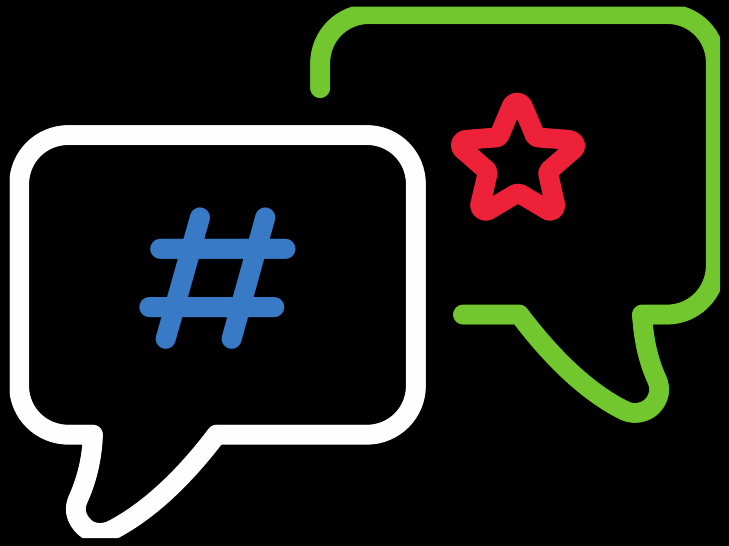
Now it’s your turn

Exercise # 3

Lookup tables



Let’s talk about: Loading



Data Target

In Data Target we define a mapping between STF and Target
In loading step, a routine can load data on different target.

Data can be stored on different format



Knowledge of Tagetik table name and fields is fundamental if target is Tagetik.
Tagetik data model can be downloaded from web application

Loading – Aggregate functions

Before

Group by					Sum
T_Sce	T_Per	T_Acc	T_Ent	.....	T_Amount
2050ACT	12	50010	A01		100
2050ACT	12	50010	A01		400
2051ACT	06	90110	B01		200
2051ACT	06	90110	B01		800

Loading – Aggregate functions

After

T_Sce	T_Per	T_Acc	T_Ent	.....	T_Amount
2050ACT	12	50010	A01		500
2051ACT	06	90110	B01		1000

- Reduces the rows to load on Target
- Improves performances

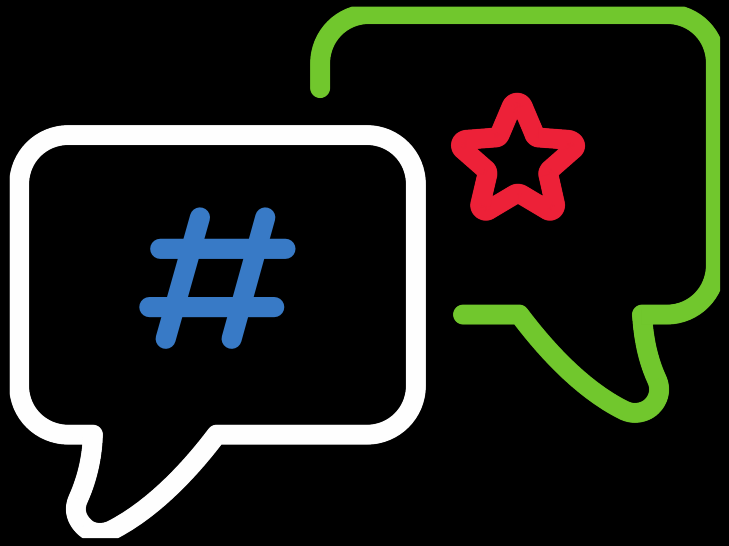
Now it’s your turn

Exercise # 4

Setup Loading



Let’s talk about: ETL Definition & Execution



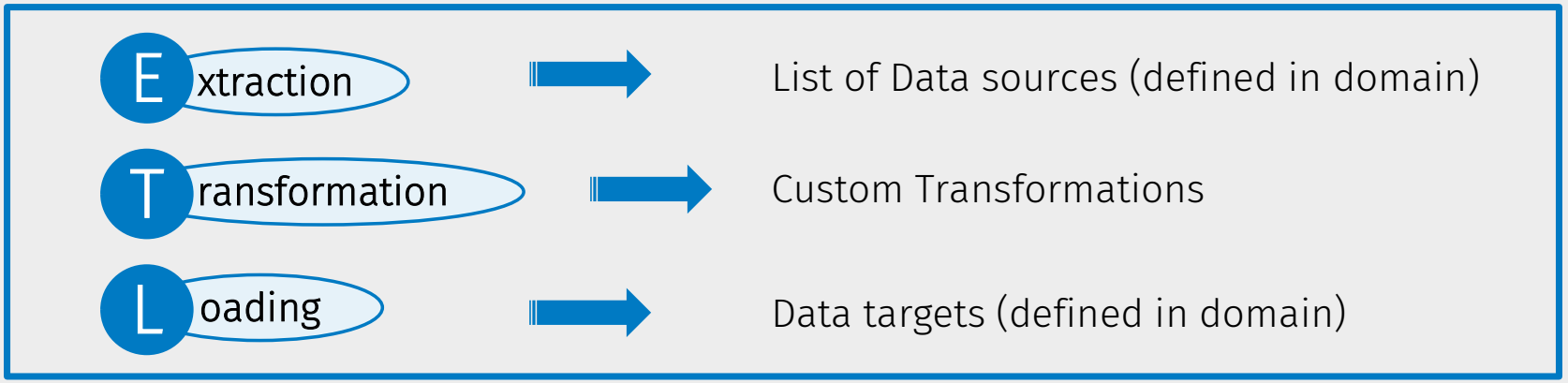
Routine definition

Smallest data processing unit in a ETL process

Combine extraction, transformation and loading steps

Add the details for Extraction, Transformation, Loading

Routine definition



Possibility to link Data Processing

Parameters definition

Store data in log tables

Available Transformations

Enrichment operations

- Update or populate fields

- Lookup table
- Formula
- Ranking
- SQL Instruction
- Matrix

Check operations

- Verify consistency of the values

- SQL Check
- Lookup table
- Dictionary
- Character
- Completeness
- Completeness and vocabulary

Others

- YTD calculation
- Verticalization
- Allocation
- Parameter choice
- Look up table feeding
- PXQ
- Delete with SQL

List of transformations

Enrichment: SQL instruction

```
T_Sce = S_Sce + 'ACT'
```

Transformations

- Set up Transformed Scenario

S_Per	S_Sce	S_Acc	S_Amount	T_Sign	T_Ent	T_Sce	T_Per	T_Acc	T_Curr	T_Amount
01	2050	AA001	100		A01	2050ACT	01			
02	2050	AA001	200		A01	2050ACT	02			
01	2050	XA001	300		A02	2050ACT	01			
02	2050	XA001	400		A02	2050ACT	02			

List of transformations

Enrichment: Lookup Table

S_Acc (Input)	T_Acc (Output)	T_Sign (Output)
AA001	50010	-1
XA001	90210	1
XA004	90240	1

Transformations

- Set up Transformed Scenario
- **Set up Transformed Account and sign**

S_Per	S_Sce	S_Acc	S_Amount	T_Sign	T_Ent	T_Sce	T_Per	T_Acc	T_Curr	T_Amount
01	2050	AA001	100	-1	A01	2050ACT	01	50010		
02	2050	AA001	200	-1	A01	2050ACT	02	50010		
01	2050	XA001	300	1	A02	2050ACT	01	90010		
02	2050	XA001	400	1	A02	2050ACT	02	90010		

List of transformations

Enrichment: SQL Instruction

```
T_Amount =    S_Amount * T_Sign
```

Transformations

- Set up Transformed Scenario
- Set up Transformed Account and sign
- Set up Amount with correct sign

S_Per	S_Sce	S_Acc	S_Amount	T_Sign	T_Ent	T_Sce	T_Per	T_Acc	T_Curr	T_Amount
01	2050	AA001	100	-1	A01	2050ACT	01	50010		-100
02	2050	AA001	200	-1	A01	2050ACT	02	50010		-200
01	2050	XA001	300	1	A02	2050ACT	01	90210		300
02	2050	XA001	400	1	A02	2050ACT	02	90210		400

List of transformations

Enrichment: SQL Instruction

```
Select COD_VALUTA
from AZIENDA
where COD_AZIENDA = T_Ent
```

Transformations

- Set up Transformed Scenario
- Set up Transformed Account and sign
- Set up Amount with correct sign
- Set up Currency

S_Per	S_Sce	S_Acc	S_Amount	T_Sign	T_Ent	T_Sce	T_Per	T_Acc	T_Curr	T_Amount_ Monthly
01	2050	AA001	100	-1	A01	2050ACT	01	50010	CAD	-100
02	2050	AA001	200	-1	A01	2050ACT	02	50010	CAD	-200
01	2050	XA001	300	1	A02	2050ACT	01	90210	EUR	300
02	2050	XA001	400	1	A02	2050ACT	02	90210	EUR	400

Verticalization

Used to verticalize consecutive column to have a proper output format on a dimension

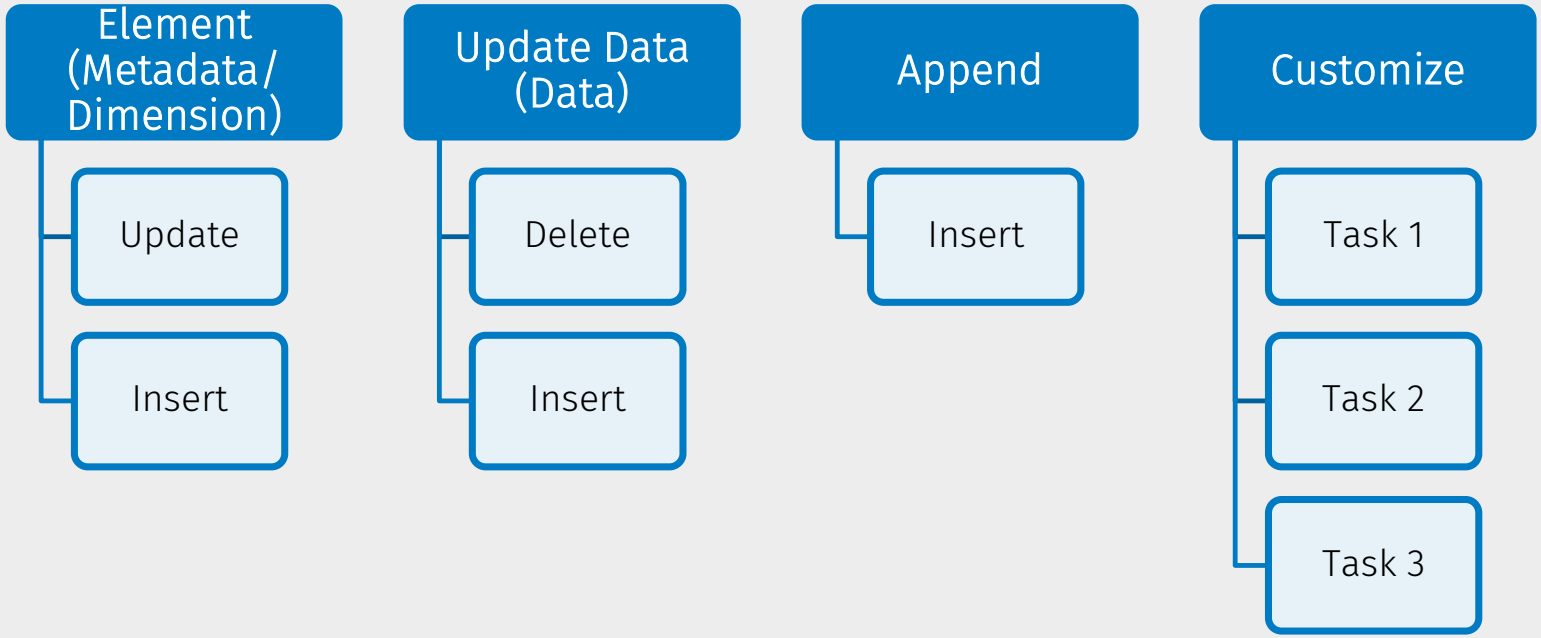
S_Scenario	S_Product	S_Price CAMPO50	S_Quantity CAMPO51	S_Revenue CAMPO52
2050ACT	A01	10	5	50
2050ACT	B01	7	3	21



T_Scenario	T_Product	T_ID	T_Amount
2050ACT	A01	CAMPO50	10
2050ACT	A01	CAMPO51	5
2050ACT	A01	CAMPO52	50
2050ACT	B01	CAMPO50	7
2050ACT	B01	CAMPO51	3
2050ACT	B01	CAMPO52	21

Loading – Loading policy

Loading policy is a list of action to be applied on target data



Options target data deletion

Using ETL filters to avoid zero or NULL value is a best practice

Use Filter Parameter to limit data to load on Target
(e.g. load only specific Entity)

`T_Ent = A02`

T_Ent	T_Sce	T_Per	T_Acc	T_Curr	T_Amount
A01	2050ACT	01	50010	CAD	-100
A01	2050ACT	02	50010	CAD	0
A02	2050ACT	01	90210	EUR	300
A02	2050ACT	02	90210	EUR	



T_Ent	T_Sce	T_Per	T_Acc	T_Curr	T_Amount
~~A01~~	~~2050ACT~~	~~01~~	~~50010~~	~~CAD~~	~~-100~~
A02	2050ACT	01	90210	EUR	300

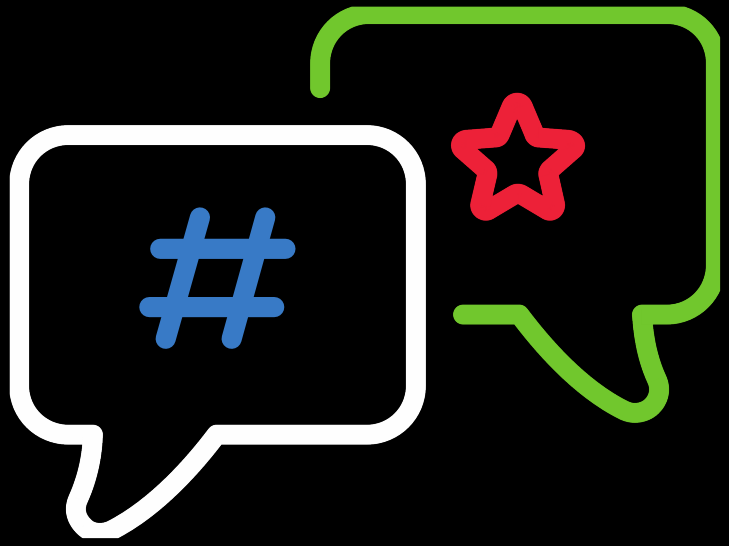
Now it’s your turn

Exercise # 5

Create ETL routine



Let’s talk about: ETL Job



Definition & Execution

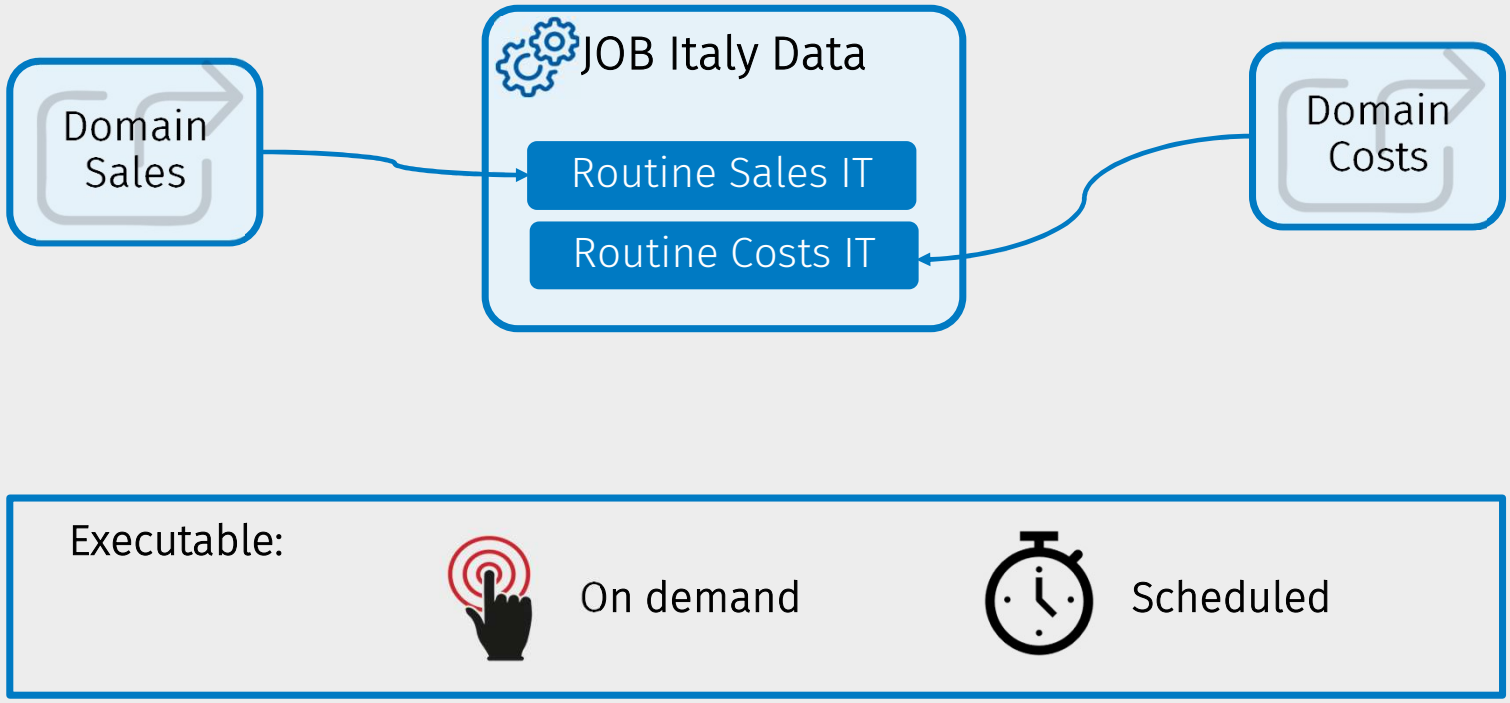
ETL Routine

- ETL Definition
- Step by step execution (troubleshooting)

ETL Job

- Faster execution not step by step (end user)
- Allows to run in sequence several routines belonging to different domains
- Possibility to reduce the interaction with user

ETL Job


Domain
Sales
JOB Italy Data
Routine Sales IT
Routine Costs IT
Domain
Costs
Executable:
On demand
Scheduled

Now it’s your turn

Exercise # 6

Create ETL job



Definition & Execution

Jobs can be executed from different menu's:

Excel and Web
Data entry form

ETL home (Admin and User interface)



Workflow

Process cockpit

Tagetik jobs

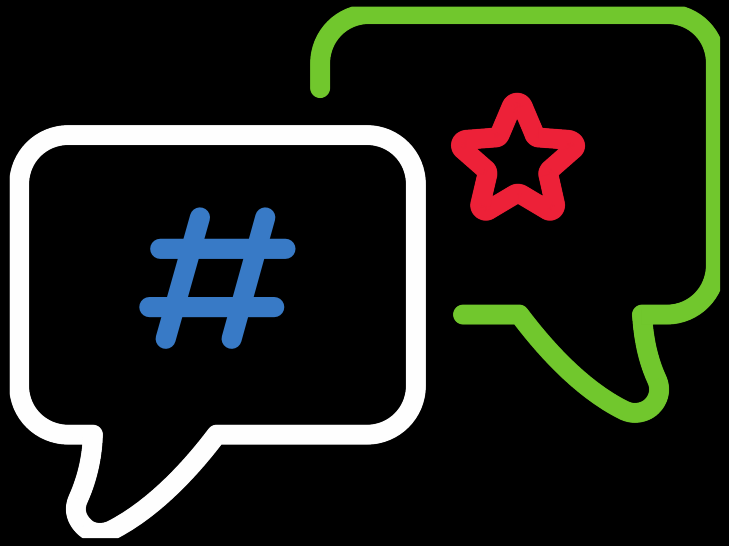
Now it’s your turn

Exercise # 7

Job execution



Let’s talk about: ETL utilities

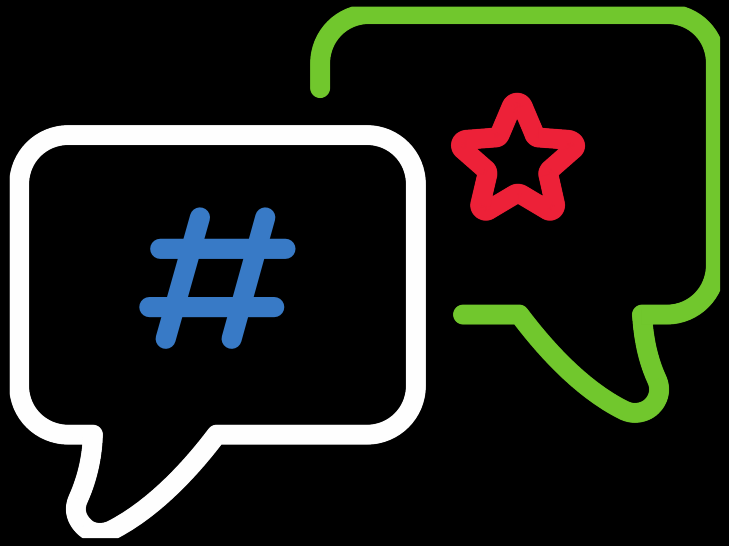


ETL utilities

Most important ETL utilities are

- Import / export ETL
- Delete ETL + staging table data
- Copy ETL

Let’s talk about: Troubleshooting & Analysis



Troubleshooting & Analysis

ETL Log Query

- Drill down to errors and warnings (Functional and Technical)

Staging Table Query

- Verify data flow on Data log tables (activate store extraction/transformation in the Routine settings)

Both can also be used for auditing

Functional Warnings/Errors

Typical Functional warnings the cause the discard of some rows:

- Dimension code does not exists
- Scenario/Period not in application process

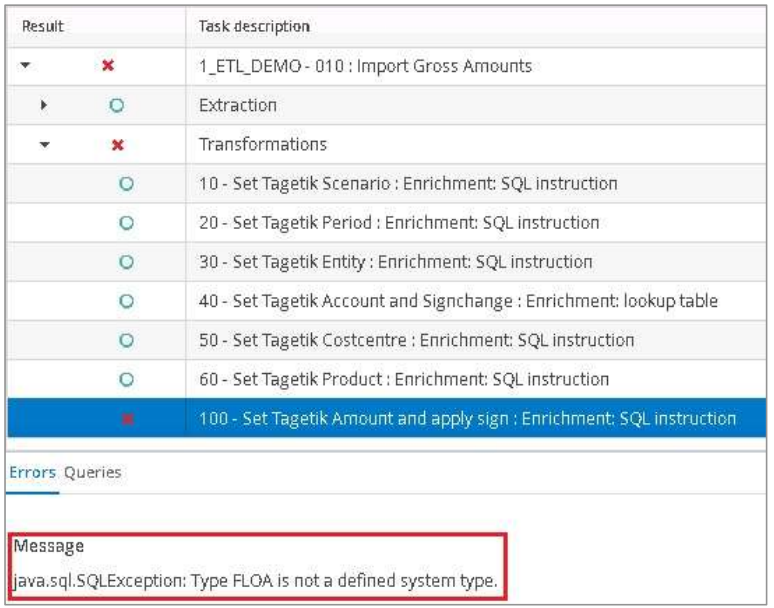


Message	1 - S_Year	2 - S_Per	3 - S_Ent	4 - S_Acc	5 - S_Cos	6 - S_Pro	10 - S_Amt	21 - S_Vers	22 - S_Src	31 - TMP_Acc_SC	51 - T_Sce	52 - T_Per	53 - T_Ent	54 - T_Acc
Entities - Code doesn't exist											2011ACT	12	E01	90210
Entities - Code doesn't exist											2011ACT	12	E01	10010
Entities - Code doesn't exist											2011ACT	09	E01	10010

Technical errors

Solve technical errors checking the pointed out ETL components!

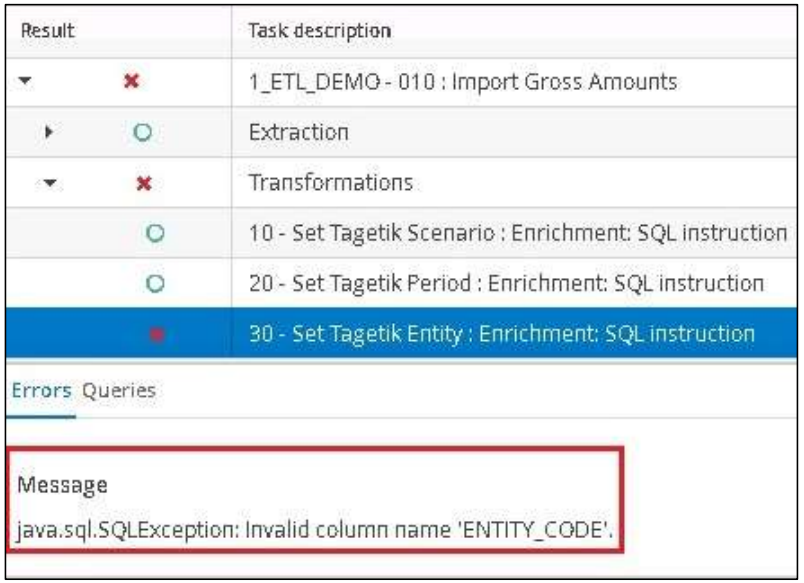




Technical errors

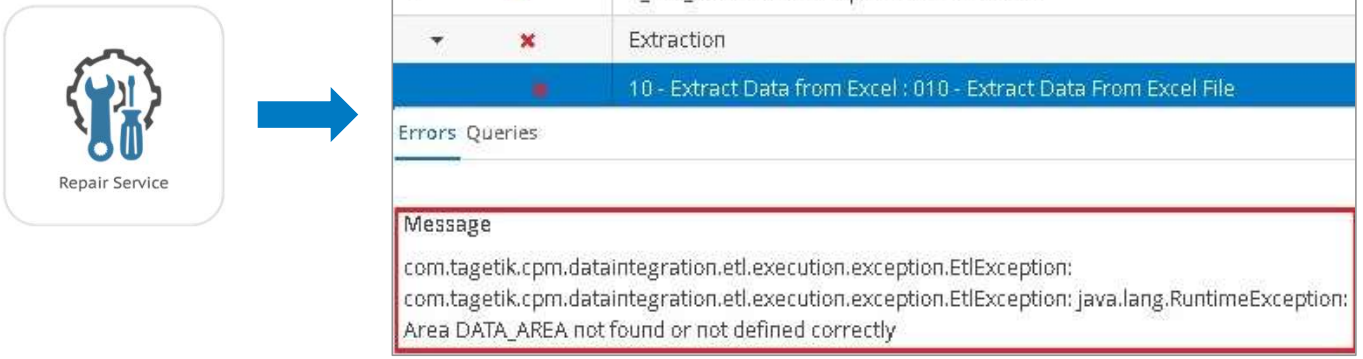
Solve technical errors checking the pointed out ETL components!





Technical errors

Solve technical errors checking the pointed out ETL components!





Takeaway

Join CCH Tagetik Insiders Hub!



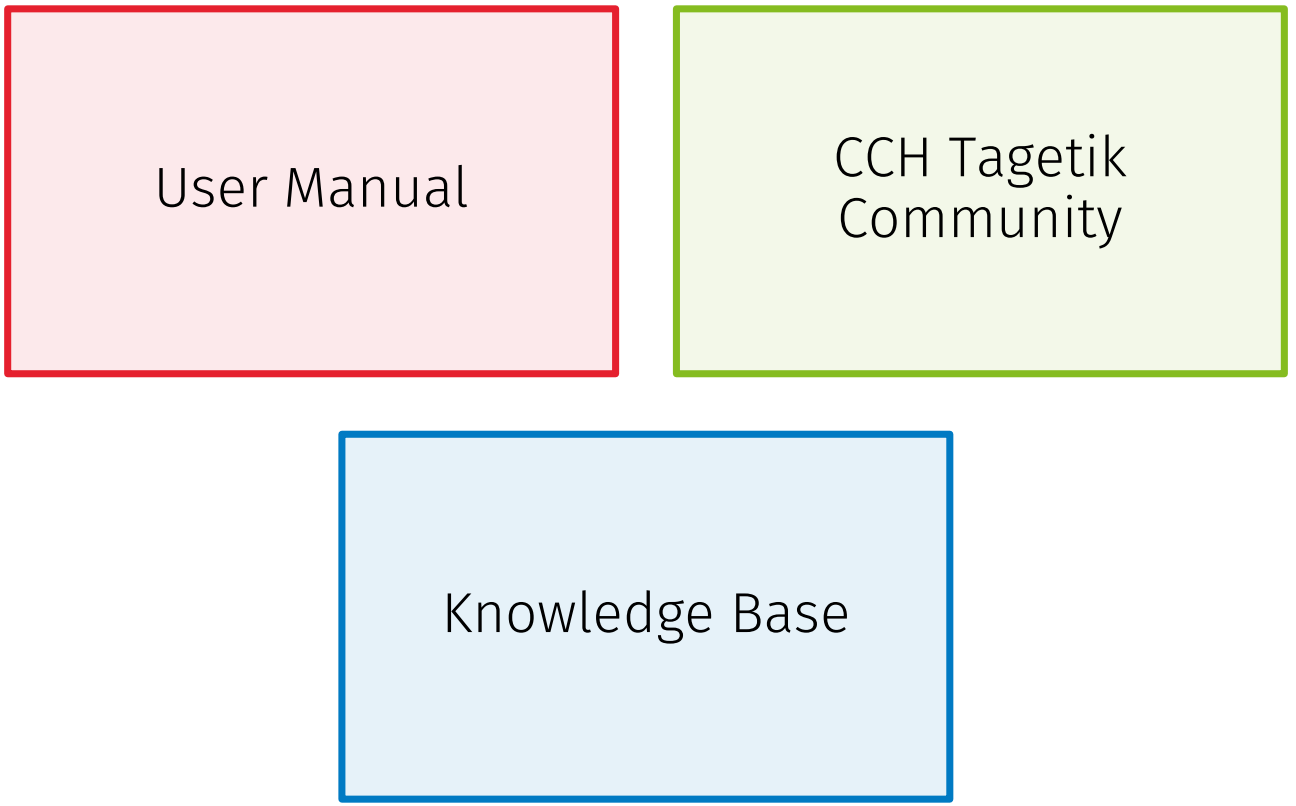
Here's the way the hub works

1. Log into the CCH Tagetik Community and click on the Insiders tab
2. Poke around and complete challenges earning points to redeem rewards
3. Get recognized for all the support you give CCH Tagetik

It's that easy!

Join CCH Tagetik Insiders now!

What other learning assets are available?



What's next?

Visit the Academy webpage

Next training and/or Assessment

Mail to info.academy@wolterskluwer.com


Thank you
Wolters Kluwer
CCH® Tagetik | ACADEMY